BangOps / Studio Operations & 24/7 Personal Agent Architecture

Bang Design · 1 Hour = 1 Credit Model · Google Workspace Native · Supabase Postgres

Technical Architecture Walkthrough

What the designer types and sees in Google Chat | **What executes in the backend**

1. Morning check-in and attendance

09:30 AM · Proactive nudge · 4 seconds total

BangOps Assistant

Morning Akash. Are you coming into the studio today?

In Studio | Working Remote | Out Sick / Leave

Selecting **In Studio** prompts active projects:

STC: BPillar | Nexa | Essent

User picks a 2.5-hour focus block. No manual forms.

Cron Runner	Scheduled job queries active team list and triggers outbound 1:1 Google Chat messages.
Jev Router	Parses button payload `{status: "in_studio", project: "STC: BPillar"}` in ~90ms.
Supabase	Updates `people.status = 'active'`. Real-time operations dashboard marks attendance.
Task Queue	Registers background timer for T + 2.5 hours to check sprint completion.

2. Sprint completion and timesheet write

12:00 PM · Timer callback · Auto-logging

BangOps Assistant

Your 2.5-hour focus block on STC: BPillar is up. What did you get done?

Finished door cap CAD modeling and added NFC clips

Assistant confirms write:
Logged 2.5 credits to `Resource_Akash_Pai.xlsx`:
"Completed door cap CAD modeling and integrated NFC sensor mount."

Groq Llama 3.1	Rewrites casual conversational note into client-ready deliverable phrasing (~300ms).
Sheets API	Authenticates via Workspace Service Account; opens user's personal spreadsheet.
Spreadsheet	Finds date row in active tab `Sep 2026`; appends `[STC: BPillar, 2.5, Log]`.
Supabase	Increments `projects.credits_burned += 2.5`. Founder dashboard updates live.

3. Drive deep-scan and vendor intelligence

02:30 PM · On-demand query · Instant benchmark

Akash (to Assistant)

Who quoted the cheapest SLA ABS 3D print for the front door panel?

Top recommendation: **Design Roots** (₹37,940, 3–4 day turnaround).
Warning: Avoid **BOTAAT** for this part. Quoted ₹81,240 (over 2x markup on large panels).

Open Quote Comparison.xlsx

Jev Router	Extracts parameters: `process: SLA ABS`, `part: front door panel`.
Drive API	Scans Google Drive; locates `3D_Print_Vendors_Quote_Comparison.xlsx`.
Supabase	Queries `vendors` table for supplier rating (4.5★) and contact details.
Gemini Flash	Identifies single-part price spread anomaly; delivers cited summary in ~550ms.

4. Unplanned absence and sub-owner failover

Anytime / 24/7 · Automatic routing · Zero project slip

Akash (to Assistant)

Sick with fever today, won't be able to come in

Assistant marks sheet `Leave = True`. Alerts sub-owner Sarah K.
Sarah receives 1:1 stand-in request:

Akash is sick on STC: BPillar. Can you cover today's CAD review?

Yes, I can cover | My plate is full

Jev Router	Detects `intent = sick_leave` with 99.4% confidence; sets status to sick.
Sheets API	Sets `Leave = True, Total Hours = 0` in Akash's monthly sheet automatically.
Failover Logic	Queries projects where `lead_owner = Akash`; identifies sub-owner Sarah K.
Escalation	Sends card to Sarah. If accepted: marks lead. If declined: alerts founders immediately.